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The Editors  
*Methods in Ecology and Evolution*  
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Dear Editors,

I am pleased to submit my manuscript, “**Auditing spatial information leakage in ecological prediction: quantifying the optimism in map accuracy with the spLeakage R package**”, for consideration as a Research Article / Application in *Methods in Ecology and Evolution*.

Ecologists increasingly turn field observations into wall-to-wall maps of species occurrence, abundance, biomass and disease risk, and judge them by an accuracy estimated through cross-validation. When data are spatially autocorrelated, ordinary random cross-validation grades a model on near-interpolation rather than on the extrapolation the map is actually used for, so reported accuracy is optimistically inflated. A mature toolkit exists to *prevent* this by building spatially separated folds (including NNDM/kNNDM and `blockCV`, published in this journal), but **no tool audits the validation an analyst has already used, or a published map, and reports how inflated the accuracy is**. The field is moreover divided over whether spatial cross-validation is even appropriate—a controversy that has left practitioners without clear guidance.

My manuscript addresses both gaps with a single idea: spatial leakage and optimism are well-defined only relative to a declared sampling *design*, *estimand* and prediction *target*. This reframing reconciles the controversy (each side is correct in a different setting) and makes optimism computable. On this basis I contribute (i) a *signed* Spatial Leakage Index that audits any split and maps where it leaks; (ii) a formal estimand for optimism under a Gaussian process, with a closed form and a Wasserstein bound; (iii) a refit-free estimator of accuracy inflation that flags when it is asked to extrapolate; and (iv) design-aware guidance, a bias-corrected accuracy that can correct a published number, and optimal validation-survey design. Everything is implemented in the open-source `spLeakage` package, which *audits rather than generates* folds and interoperates with the existing ecosystem.

The work is a strong fit for *MEE*: it builds on and complements methods this journal has championed (NNDM, the area of applicability, `blockCV`), pairs a methodological advance with documented open-source software, and speaks to emerging reporting standards for distribution models. Key results: against exact Gaussian-process algebra the index is a near-sufficient statistic for true optimism ( $r = 0.98$ ) and recovers the *direction* of bias that unsigned matching statistics cannot ( $r = 0.98$  vs  $-0.09$ ); auditing 16 real datasets, reported accuracy is optimistic in *every* case (median 22%); and two contrasting case studies—a species distribution model and a malaria-prevalence map—show the audit correctly attributing leakage to genuine autocorrelation in one and to a survey-design artefact in the other.

The manuscript is original, is not under consideration elsewhere, and has not been published previously; there are no conflicts of interest. The `spLeakage` package and all analysis code are openly available at <https://github.com/olatunjijohnson/spLeakage> (documentation at <https://olatunjijohnson.github.io/spLeakage/>); all datasets are public, and code will be archived with a DOI on acceptance. I have suggested several potential referees spanning both the spatial cross-validation and design-based map-validation perspectives the manuscript reconciles. Thank you for considering my submission.

Yours sincerely,  
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